POWER PULSE

Circuit Breakers Important Formulae

Notes by BALAJI SIR

1 Ratings

1. $$\text{Breaking MVA} = \sqrt{3}\, V_L(\text{kV})\, I_b(\text{kA}) \quad \text{(symmetrical rms at rated voltage)}$$
2. $$\text{Making current} \approx 2.55 \times I_{b,sym}\ \text{(peak)}, \qquad I_{sc} = \frac{MVA_{base}}{\sqrt{3}\, V\, X_{pu}}$$
3. $$\text{Rated sequence: } O - 0.3\ \text{s} - CO - 3\ \text{min} - CO$$

2 Transient recovery voltage (TRV)

4. $$\omega_n = \frac{1}{\sqrt{LC}}, \qquad f_n = \frac{1}{2\pi\sqrt{LC}}$$
5. $$v(t) = E_m\,(1 - \cos \omega_n t), \qquad v_{peak} = 2E_m$$

   E_m = peak phase voltage (undamped)
6. $$RRRV_{max} = E_m\, \omega_n \quad (\text{at } \omega_n t = \tfrac{\pi}{2})$$
7. $$R_{critical} = \frac{1}{2}\sqrt{\frac{L}{C}} \quad \text{(resistance switching: no oscillation for } R \le R_c)$$

   parallel resistor across contacts

3 Overvoltages

8. $$\tfrac{1}{2}LI^2 = \tfrac{1}{2}CV^2 \;\Rightarrow\; V = I\sqrt{L/C}$$

   current chopping (inductive switching)
9. Capacitive switching: restrike can give up to $3\times$ the peak voltage

4 Types

10. ACB, oil (bulk/minimum), air-blast, vacuum, SF_6 (best for EHV)
11. Arc extinction at current zero: high-pressure cooling, high resistance, high-dielectric gas